Experiment-12

Write A Python Program For Tic Tac Toe Game

Aim

To write and execute a Python program to implement the Tic-Tac-Toe game for two players, where the first player to get three marks in a row, column, or diagonal wins.

Algorithm

1. Start the program.
2. Initialize a 3 × 3 game board with empty spaces.
3. Assign X to the first player and O to the second player.
4. Display the game board.
5. Ask the current player to enter a position from 1 to 9.
6. Check whether the entered position is valid.
7. Check whether the selected position is empty.
8. Place the player's mark in the selected position.
9. Check whether the player has three marks in a row, column, or diagonal.
10. If a player wins, display the winner and stop the game.
11. If the board is full and there is no winner, declare a draw.
12. Otherwise, switch players and repeat the process.

Code

ex-12.py - D:/ex-12.py (3.10.0)

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```
# Tic-Tac-Toe Game

board = [" "] * 9

def display_board():
    print()
    for i in range(0, 9, 3):
        print(" " + " | ".join(board[i:i + 3]))
        if i < 6:
            print("---+---+---")
    print()

def check_winner():
    wins = [
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)
    ]

    for a, b, c in wins:
        if board[a] == board[b] == board[c] != " ":
            return board[a]

    return None

def play_game():
    player = "X"

    print("TIC-TAC-TOE GAME")
    print("Positions are numbered 1 to 9.")
    print("Player X and Player O take turns.")

    while True:
        display_board()
```

```
        try:
            position = int(input(
                "Player " + player + ", enter position (1-9): "
            )) - 1
        except ValueError:
            print("Please enter a number.")
            continue

        if position < 0 or position > 8:
            print("Invalid position! Try again.")
            continue

        if board[position] != " ":
            print("Position already occupied! Try again.")
            continue

        board[position] = player

        winner = check_winner()

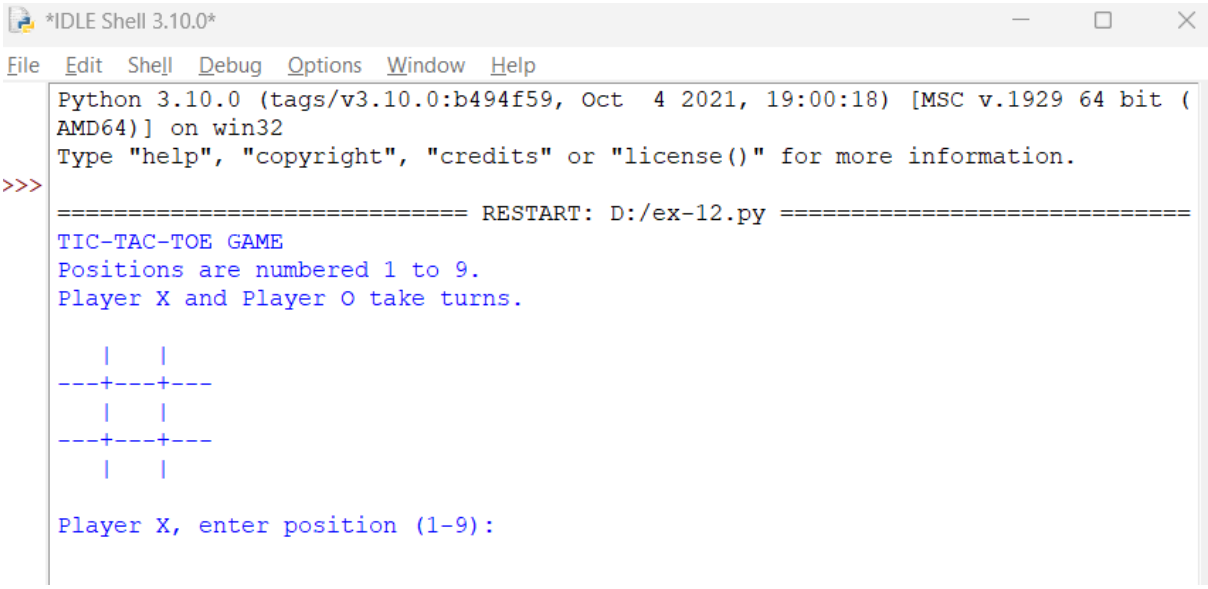
        if winner:
            display_board()
            print("Player", winner, "wins!")
            break

        if " " not in board:
            display_board()
            print("Game is a draw!")
            break

        player = "O" if player == "X" else "X"

play_game()
```

Output



Result

The Tic-Tac-Toe game was successfully implemented. The program allows two players to take turns, checks for a winner, and declares a draw when the board is full without a winner.